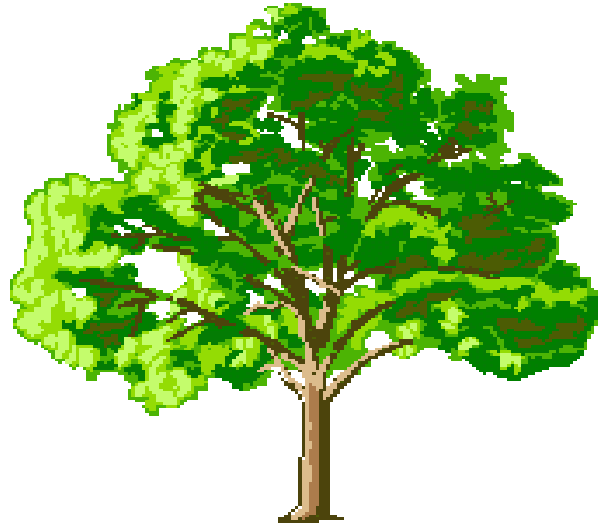


B+ Trees



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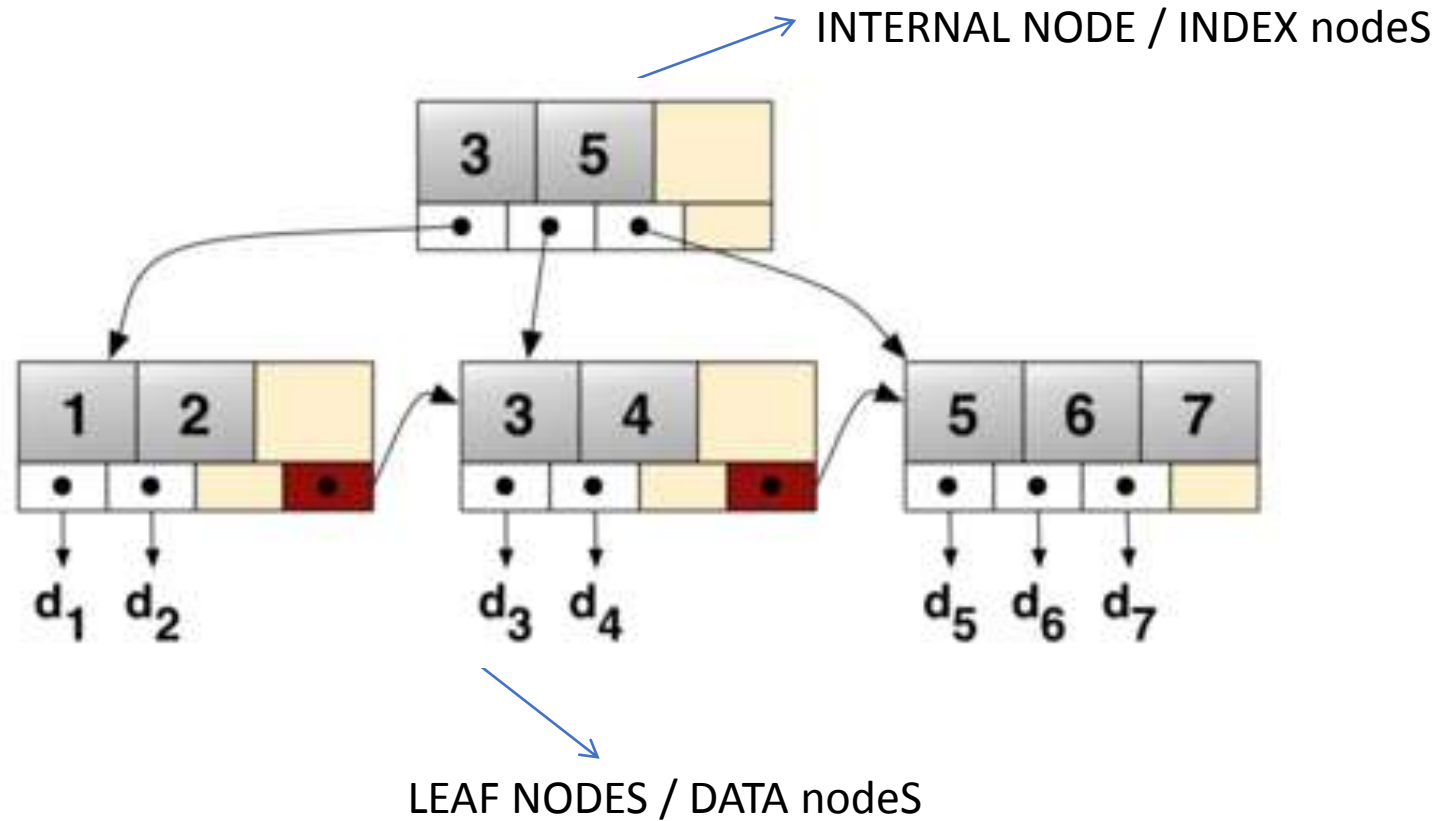
What is b+ tree

- ❑ A B+ tree is a data structure often used in the implementation of database indexes. Each node of the tree contains an ordered list of keys and pointers to lower level nodes in the tree.
- ❑ A B+ tree is a balanced tree in which every path from the root of the tree to a leaf is of the same length.

B+- Tree Characteristics

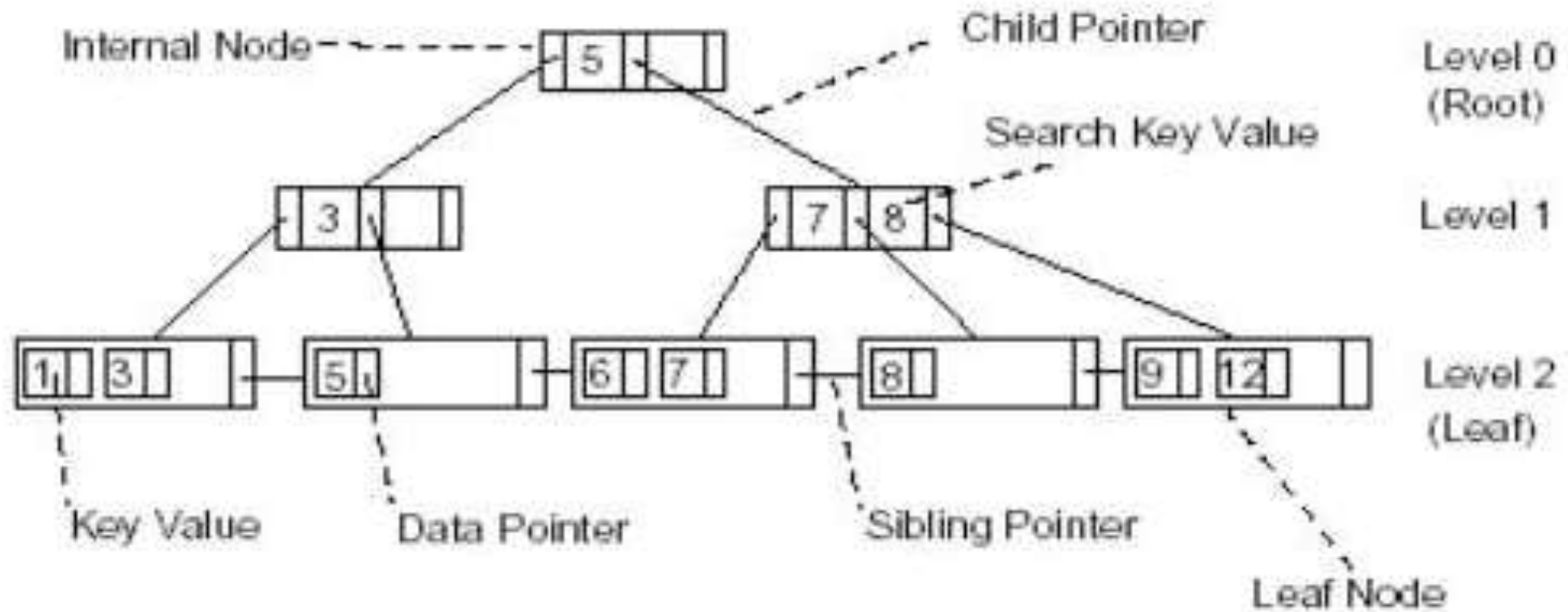
- Data records are only stored in the leaves.
- Internal nodes store just keys.
- All data is stored at the leaf nodes (leaf pages); all other nodes (index pages) only store keys.
- All the leaf nodes are interconnected with each other leaf nodes for faster access.
- Keys are used for directing a search to the proper leaf.
- If a target key is less than a key in an internal node, then the pointer just to its left is followed.
- If a target key is greater or equal to the key in the internal node, then the pointer to its right is followed.
- B+ Tree combines features of ISAM (Indexed Sequential Access Method) and B Trees.

B+ TREE



The linked list allows rapid traversal.

Structure of b+ tree



How to create or insert in B+:

=2,4,7,10,17,21,28

if $n=4$ n is the order

$\text{key} = n - 1 = 4 - 1 = 3$

2		
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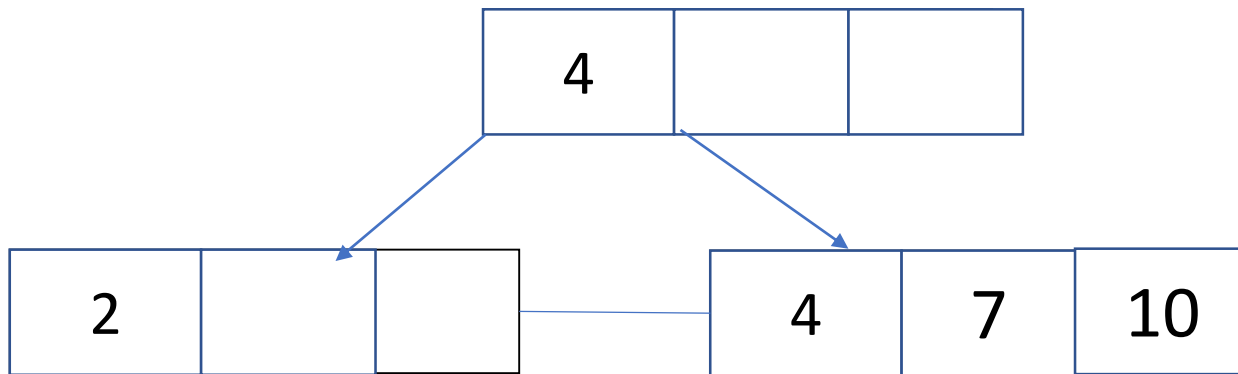
$=2, 4, 7, 10, 17, 21, 28$

2	4	
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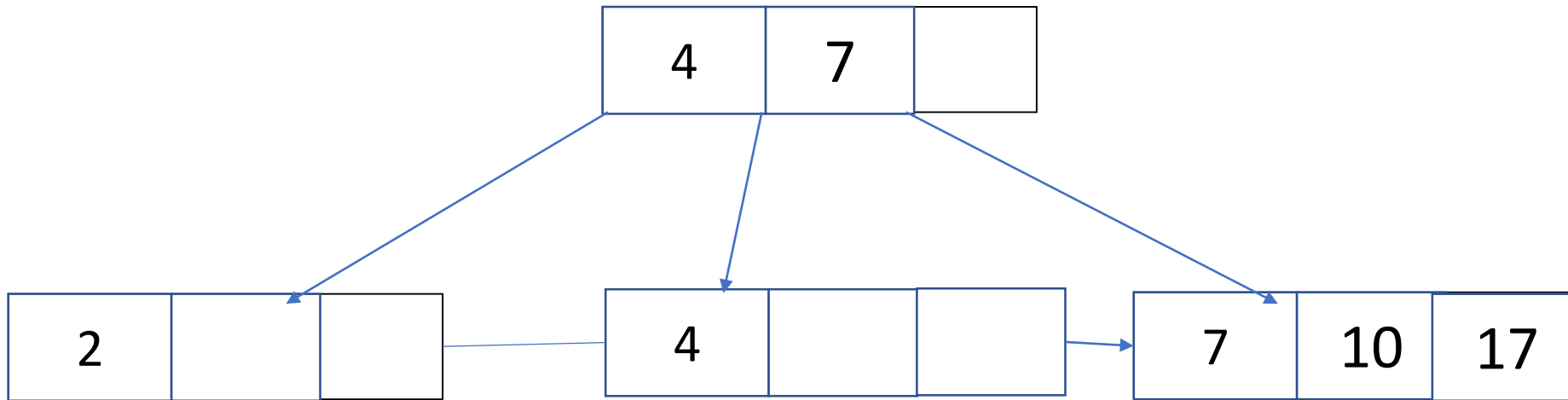
$=2, 4, 7, 10, 17, 21, 28$

2	4	7
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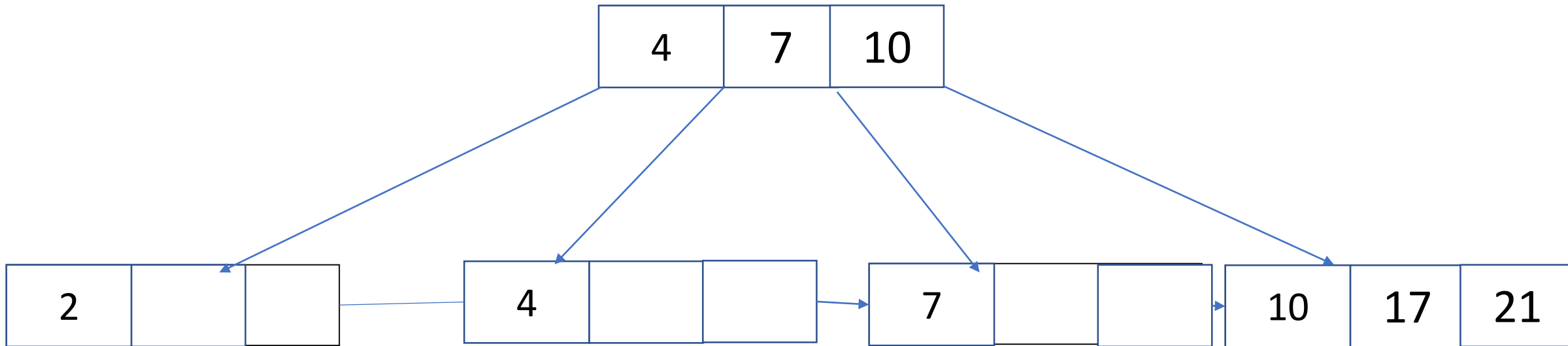
$=2, 4, 7, 10, 17, 21, 28$



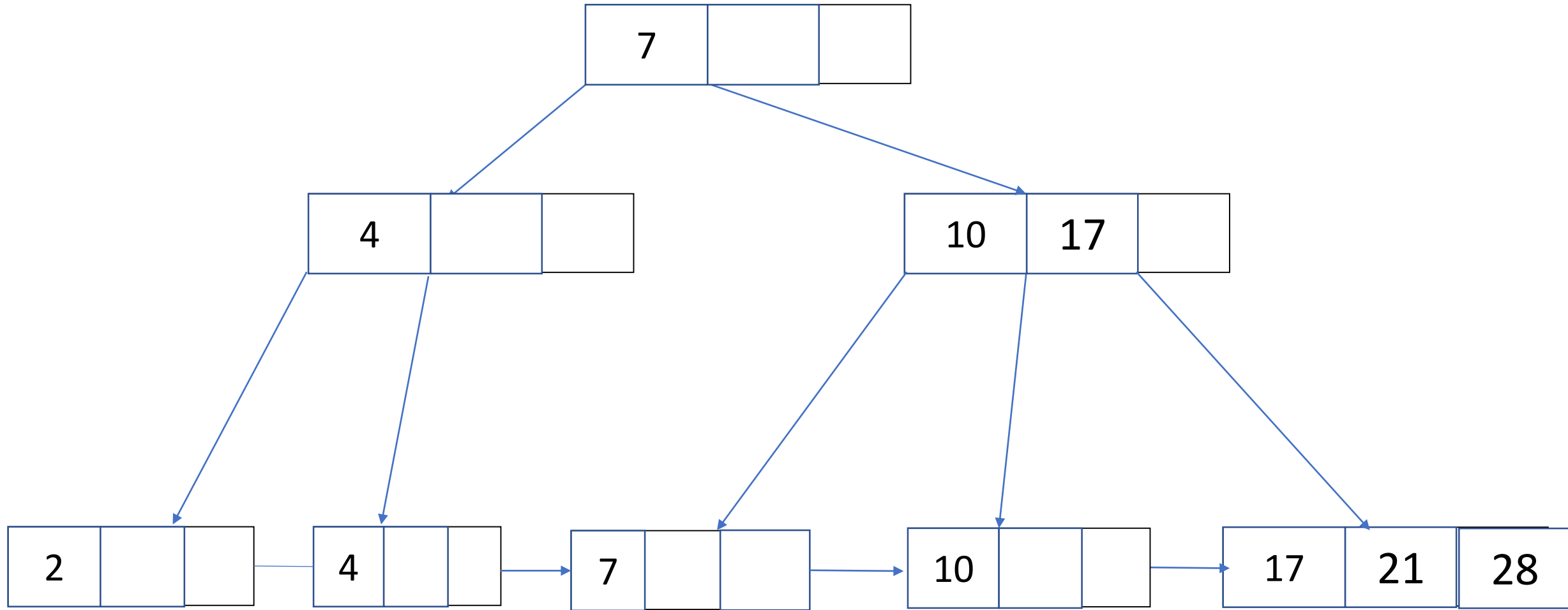
==2,4,7,10,17,21,28



$=2, 4, 7, 10, 17, 21, 28$

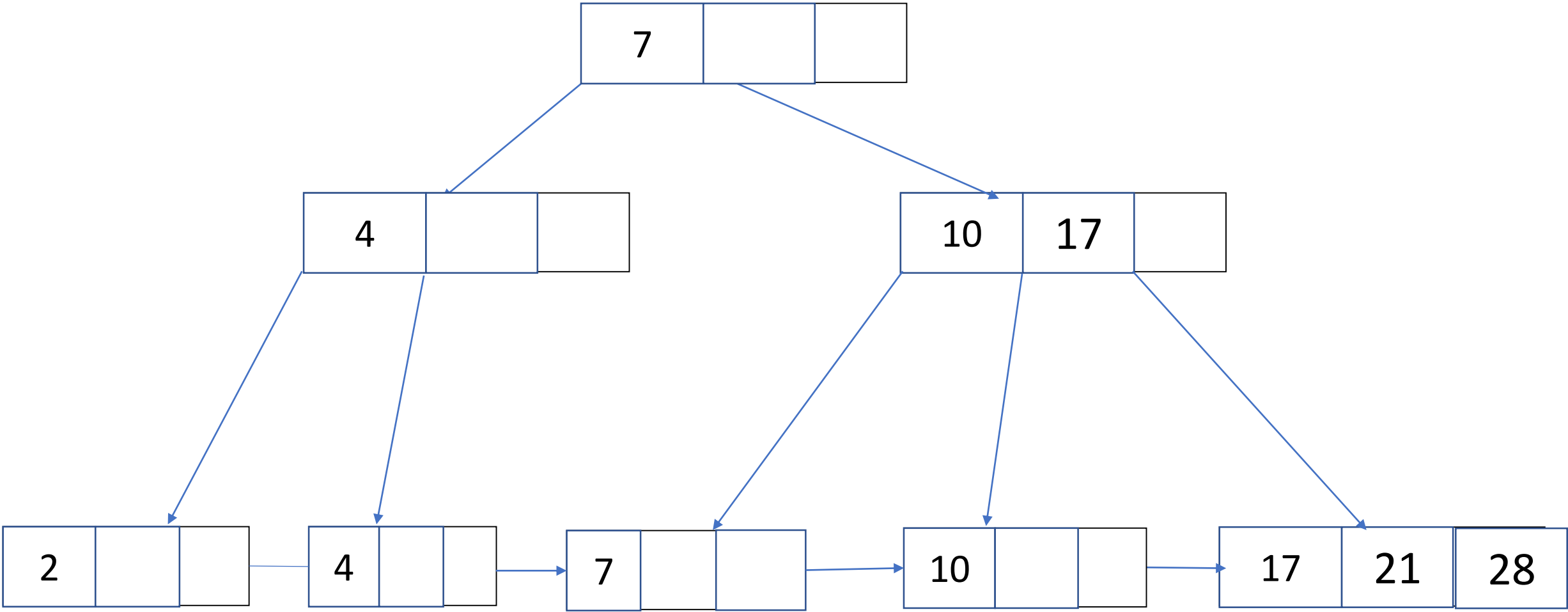


==2,4,7,10,17,21,28

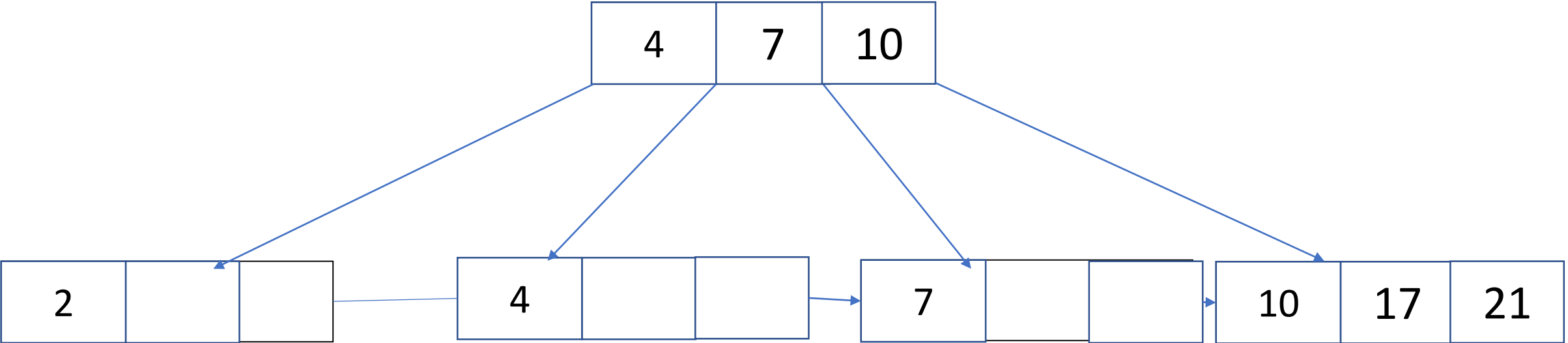


Deletion Process:

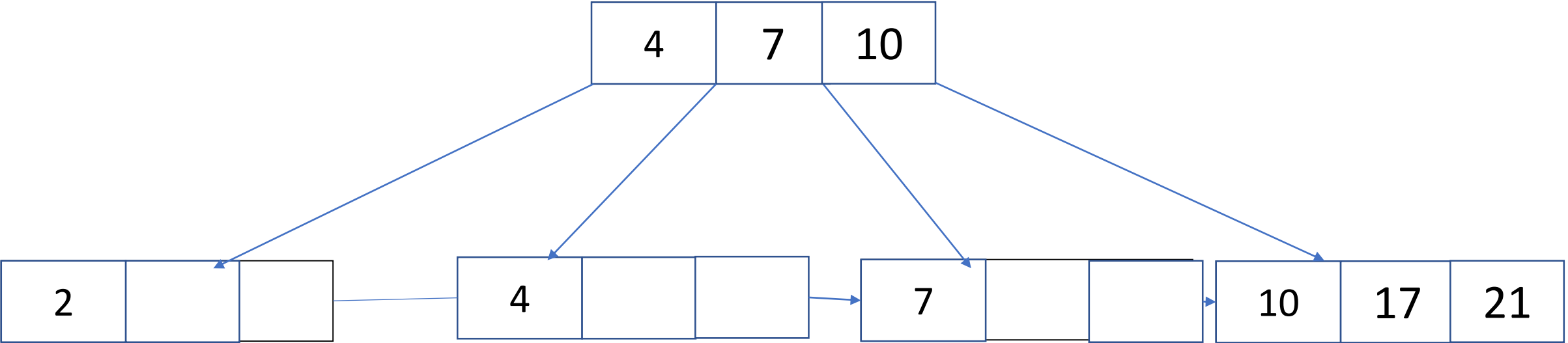
Deletion of 28:



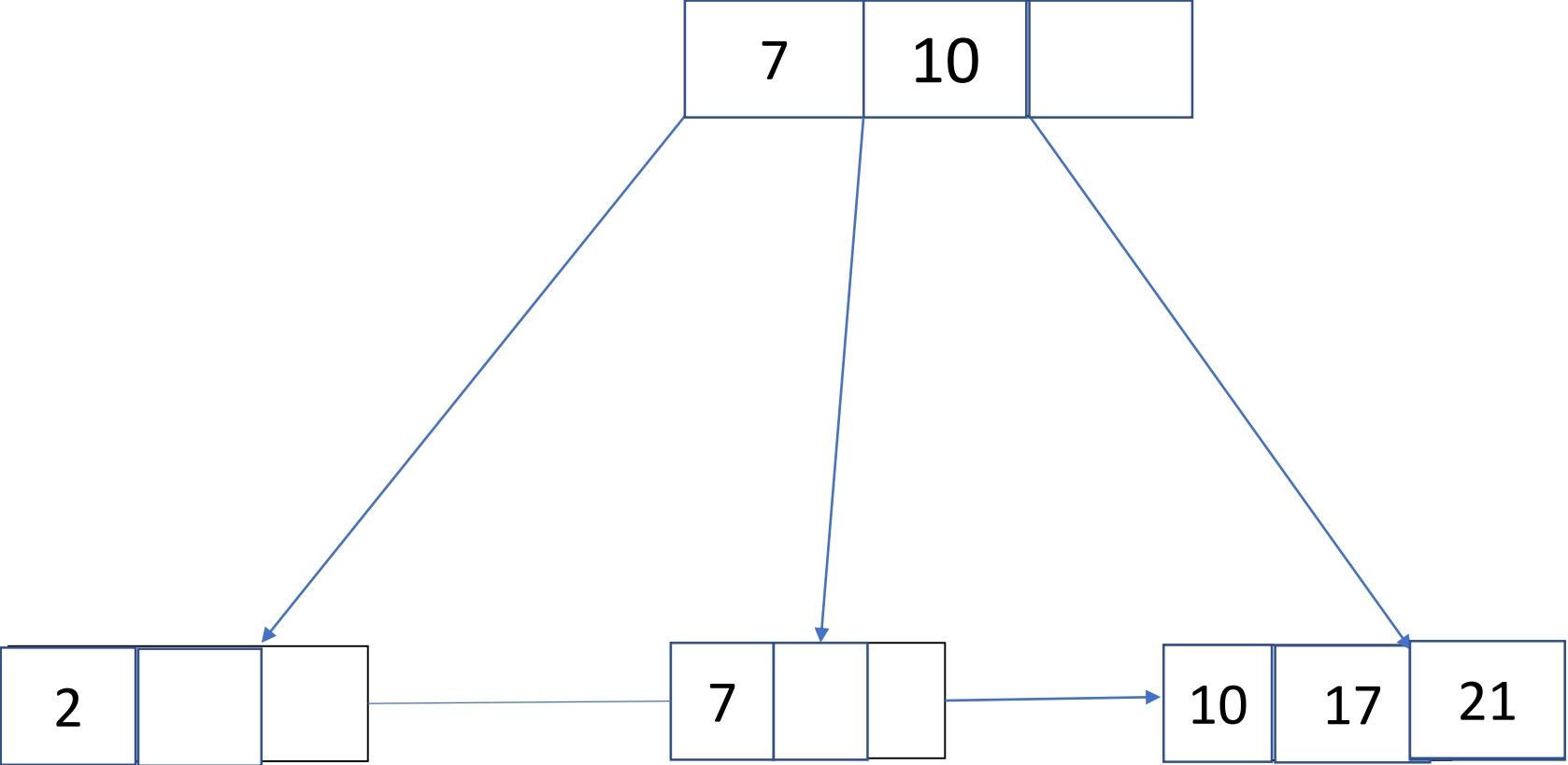
Deleted 28:



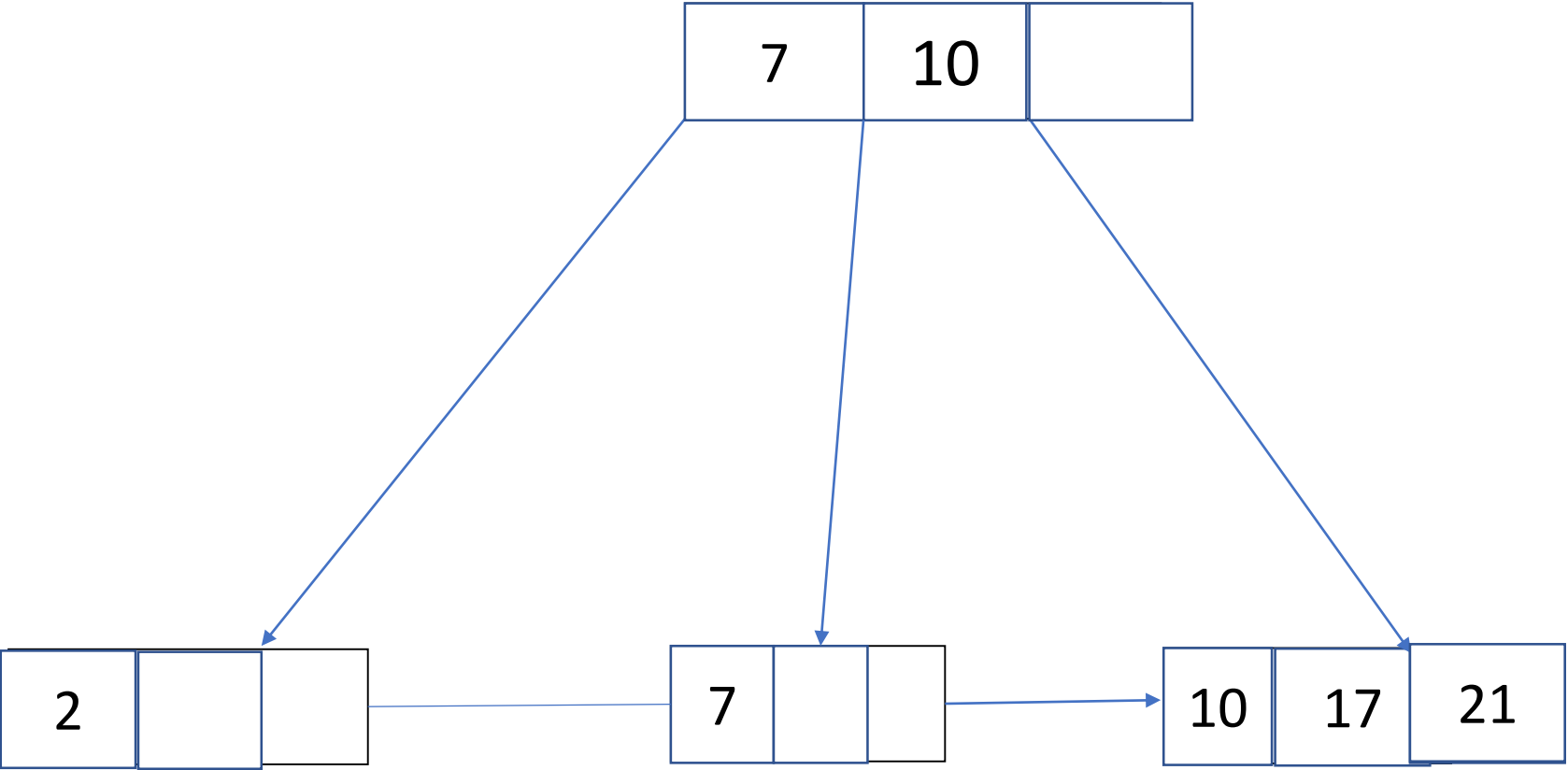
Deletion of 4:



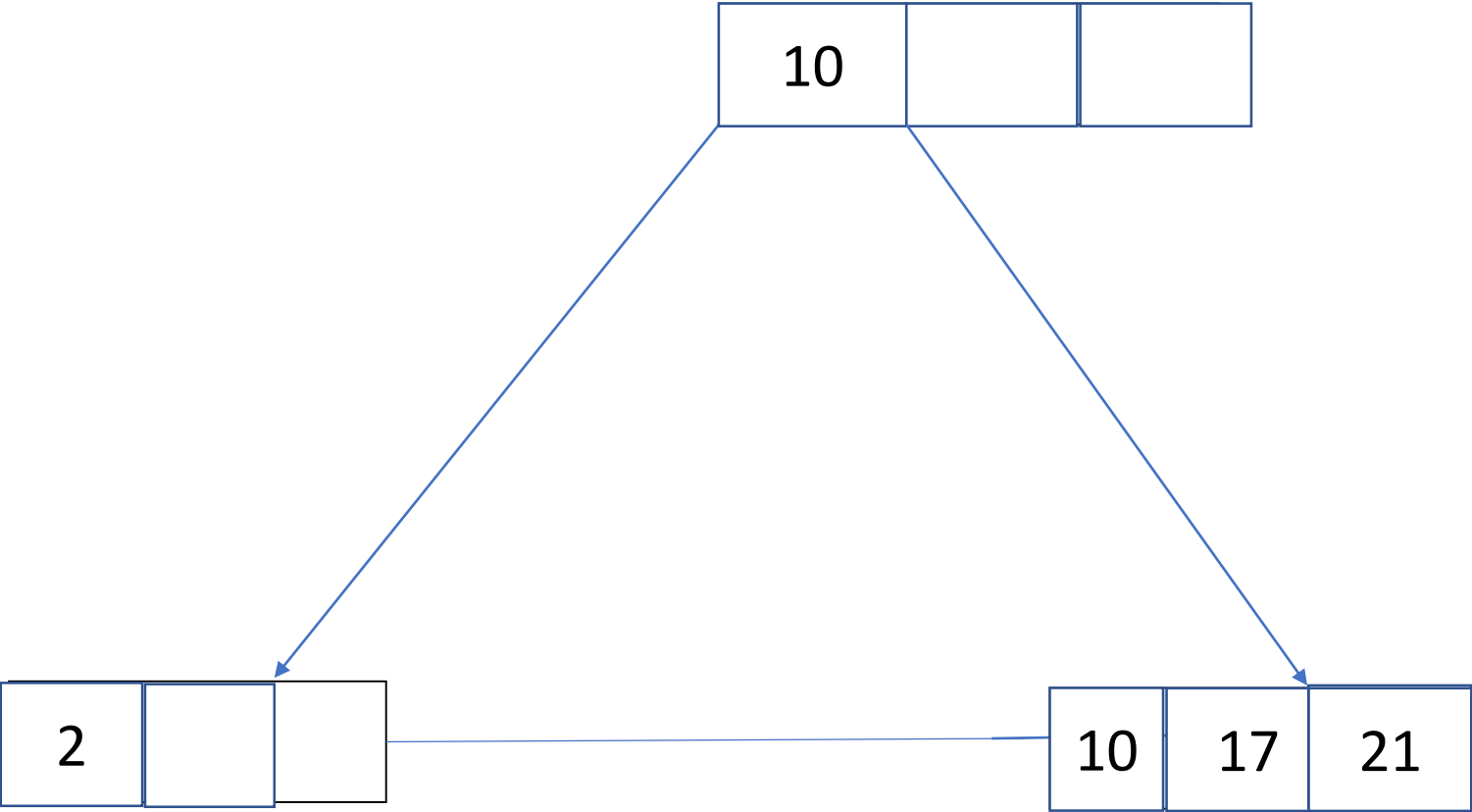
Deleted 4:



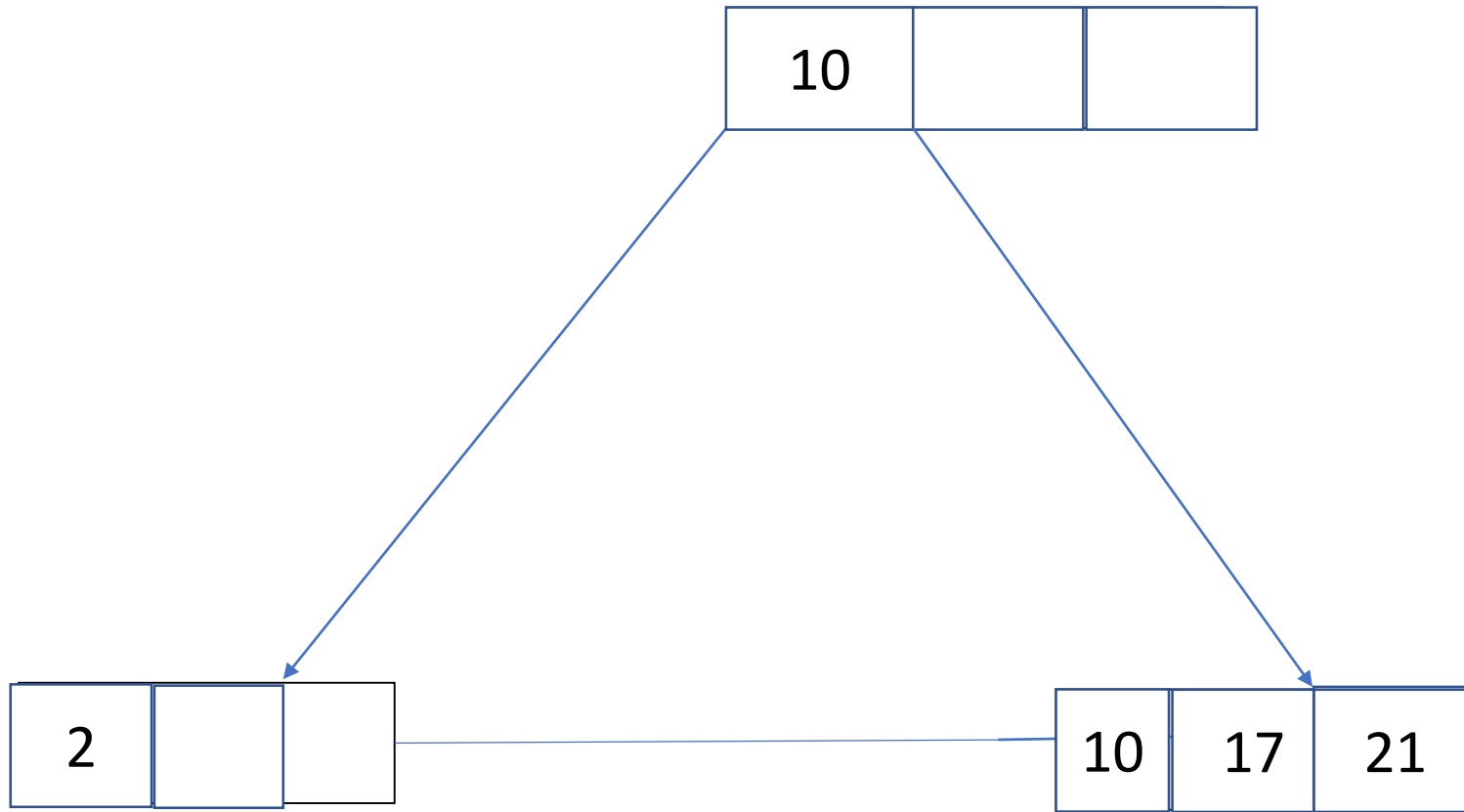
Deletion of 7:



Deleted 7:



Deletion of 2:



Deleted 2:

10	17	21
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Thank you